

When the Baseline Also Abstains: Pitfalls in Evaluating Abstention for Document Audit

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Preprint — June 2026

Abstract

Retrieval-augmented (RAG) auditors invent disagreements: asked for a value the document never gives, they answer anyway, producing **false-positive divergences**. We build an **ontology-gated** auditor that abstains on type/scope mismatch—and, more importantly, we find that *measuring* whether such a system abstains well is itself error-prone. Using a benchmark of Brazilian contracts and government tender notices, we surface **three pitfalls**, each illustrated by a concrete mistake we made and corrected: (i) a **single-sample illusion**, where a rare false-positive rate reads as “0%” until bootstrapped, and a self-consistency “free lunch” turns out to be sampling noise; (ii) that **“hard” is defined by the model, not the taxonomy**—a naive baseline ties an abstaining system not only on wholly-missing concepts but even on taxonomically-adjacent ones (an engineering benchmark we built to discriminate did not), so a gate helps only where the model actually conflates a concept pair; and (iii) a **keyword-absence annotation trap**, where labeling ABSTAIN from a missing keyword is invalid on real legal text that paraphrases concepts. We distill the lessons into a checklist for evaluating abstention in document audit, and release the system, benchmark, and harnesses.

1 Introduction

Organizations compare many structured-but-unstructured documents—contracts, tender notices (*editais*), financial statements, engineering specification sheets—to find discrepancies. A natural approach is RAG: retrieve passages and ask an LLM to extract the answer per document, then compare. This fails in a specific, costly way: **standard RAG prefers answering to abstaining**. Asked “what is the penalty for *total non-performance*?” of a contract that only sets a *late-payment* penalty, a naive extractor returns the late-payment value—fabricating an answer to a question the document never addresses. Across documents these become **false-positive divergences**: flagged disagreements that are artifacts of hallucination, and in auditing the costliest error.

The obvious fix is to make the system abstain. We do (Section 3), and evaluate it across **two domains**—Brazilian legal contracts and tender notices (Portuguese), and English reliability and mechanical engineering specification sheets—so the lessons do not hinge on one domain. But our main finding is methodological: **whether an abstaining system is actually better is surprisingly hard to measure**, and the natural ways to measure it quietly mislead. We report three pitfalls, each a mistake we made first and caught later:

- **Pitfall A — single-sample illusion** (§4.2). Our first runs showed “ $0\% \pm 0\%$ ” false positives; bootstrapping revealed $\sim 1\text{--}2\%$. The zero was sampling noise, and it made a permissive self-consistency vote look like free recall.

- **Pitfall B** — “hard” is the model’s, not the taxonomy’s (§4.3). A baseline ties our gate not only where a concept is wholly absent but even on taxonomically-adjacent pairs (a 21-case engineering benchmark we built to discriminate did not). A gate helps only where the *model* conflates a pair.
- **Pitfall C** — **keyword-absence annotation trap** (§4.4). We labeled ABSTAIN from a missing keyword; three labels were wrong because the concept was present under another name. The gate had answered correctly; our benchmark was lying.

Contributions. (1) An ontology-gated auditor with explicit abstention and a deterministic comparison judge (case study, Section 3). (2) Three evaluation pitfalls for abstention in document audit, each demonstrated on a real failure (Section 4). (3) A checklist for evaluating abstention, plus a released benchmark and harnesses.¹

2 Related Work

RAG and its hallucinations [10, 3]; our concern is answering despite absent in-scope evidence. **Structured extraction** (function-calling, Instructor [7], Guardrails [5], constrained decoding [14]) coerces typed outputs; we make the schema’s primary job an abstention pre-condition. **Selective prediction / abstention** [4, 9] and answerability [12] are the closest neighbors; we study how to *evaluate* abstention in multi-document audit. **Faithfulness and RAG evaluation** [8, 11, 2] ask whether an answer is supported; we target the dual question—whether a system correctly declines—and how its benchmarks mislead. **Legal NLP** [6, 1] supplies the domain.

3 The Case-Study System

FLARE26 is a RAG pipeline used as a vehicle for the evaluation study: hybrid retrieval (M1.5) → ontology-gated extraction (M2) → deterministic N -way judge (M4) → summary (M5), with provenance to the source span.

Ontology-gated extraction (M2). A typed record with two independent checks: *type/instrument* (the named instrument—e.g. a *penalty* vs. *interest*, distinct even when both are percentages; must match) and *scope/condition* (strict equality). If either fails, the record collapses to an explicit *evidence gap*. The rule is **domain-agnostic** (stated over abstract instruments P/Q , conditions A/B ; no hard-coded legal terms).

Deterministic N -way judge (M4). Groups documents by equivalent answer (BR number normalization: R\$ 1.200.000,50 → 1200000.5; 12,5% ≡ 12.5); verdict *consensus* / *divergence* / *evidence gap* in $O(N)$, no LLM call. An upstream hallucination is the only way to fabricate a divergence—what the gate suppresses. **Optional self-consistency:** run k times and vote; the threshold t is a precision/recall knob (§4.2).

4 Evaluation and Three Pitfalls

Benchmark. 61 verified (question, document) pairs across two domains and two languages: a **legal** split—30 synthetic Brazilian contracts (13 ABSTAIN, 17 answerable) and 10 real tender notices (4 ABSTAIN, 6 answerable), in Portuguese—and an **engineering** split—21 English reliability/mechanical spec sheets (12 ABSTAIN, 9 answerable), with both *type* near-neighbors (MTBF

¹Code, benchmark, and evaluation harnesses: <https://github.com/marcospaula/FLARE26-P1>

vs. MTTR, tensile vs. yield) and *scope* near-neighbors (same metric, different condition: tensile at 20 °C vs. 200 °C; torque for bolt class 8.8 vs. 10.9). Extractor: gpt-4o-mini, temperature 0, fixed seed. Metrics: false-positive rate (fraction of ABSTAIN answered), abstention recall, answer recall.

4.1 The synthetic result (what we wanted to report)

System	False-positive ↓	Answer recall ↑
Baseline (free extraction)	38% ± 0%	100% ± 0%
FLARE26 (gated)	~1–2%	81% ± 6%

Ontological gating cuts false-positive divergences ~20–30×. Ablation: relaxing the strict scope rule to accept sub/superset conditions recovers recall but **reopens** false positives (0% → 38%), so the strict rule is the operating point.

4.2 Pitfall A — the single-sample illusion

Our first measurement reported 0%±0% false positives, and a vote “answer if ≥ 1 of k ” appeared to recover recall *for free*. Both were artifacts of measuring a rare event once. Bootstrapping the per-item sample pool: The gate leaks rarely (~1–2% per sample), and “ ≥ 1 ” *amplifies* it with k .

k (LLM calls/doc)	False-positive	Answer recall
1	1% ± 2%	81% ± 6%
5	3% ± 4%	91% ± 3%
10	5% ± 4%	93% ± 2%

Sweeping the vote threshold t shows “ ≥ 1 ” was the wrong knob [13]; a majority-style t filters the leaks: (The two tables are independent bootstrap runs; the small disagreement at $k=1$ —1% vs. 2%

Policy	False-positive	Answer recall
single call ($k=1$)	2%	77%
$k=10, t=1$ (“ ≥ 1 ”)	10%	86%
$k=10, t=3$	1%	82%
$k=10, t=4$	0%	80%

FP, 81% vs. 77% recall—is itself the sampling noise this section is about.) The point $k=10, t=4$ dominates the single call on both axes. **Lesson:** report rare-event metrics as mean ± std over repeated runs; sweep the vote threshold; never trust a single “0%”.

4.3 Pitfall B — “hard” is defined by the model, not the taxonomy

The legal 38% baseline false-positive rate comes from absences where a *similar-but-wrong concept is present*—asked for a *penalty*, the document has *interest*. It is tempting to call these “hard” because the concepts are taxonomically adjacent; they are hard rather because **the model conflates the pair**. Two controls make this precise.

Wholly-absent concepts are easy—even for the baseline. Our first real ABSTAIN cases had the concept wholly missing; retrieval returns nothing and **even the baseline abstains**, tying

the gate at 0%. A 10-pair real pilot (4 such absences, 6 answerable) gives FLARE 0% / 83% recall vs. baseline 0% / 50%: the gate wins on recall, the false-positive comparison is uninformative.

Taxonomically-adjacent $\neq$ hard. We built 21 engineering cases with *type* near-neighbors (MTBF vs. MTTR, tensile vs. yield) and *scope* near-neighbors (20 °C vs. 200 °C; class 8.8 vs. 10.9). We expected the baseline to leak; **it did not**—on all 12 engineering ABSTAIN cases it abstained correctly (0% false positives), matching the gate. The model does not confuse these crisp concepts. A control isolates the cause: the legal *interest-for-penalty* leak persists when the case is translated to English, so it is the **concept pair**, not the language or domain, that the model treats as interchangeable.

Consequences. An abstention gate helps *only where the baseline leaks*, which depends on the model’s concept-confusability—not the annotator’s taxonomy, the domain, or the language. In crisp domains the baseline is already careful, so the gate does no harm (the engineering split: 0% false positives, 100% recall) but adds nothing. **Lesson:** establish concept-difficulty empirically (does the baseline leak?), not by assuming adjacency implies it.

4.4 Pitfall C — the keyword-absence annotation trap

We labeled real-document ABSTAIN from a missing *keyword*. Three labels were wrong, because legal text paraphrases concepts: *juros de mora* appeared as *compensação financeira*; “payment term” as “payment within 30 days”. In each case the gate had **answered correctly** and our label called it a false positive. Re-annotating absence by **concept** (exhausting synonyms before declaring ABSTAIN) corrected them; one ambiguous case is held out pending domain adjudication. **Lesson:** never infer absence from a missing keyword on real text; annotate by concept, with a second annotator.

4.5 Separating retrieval from the gate

Diagnosing a real-document recall failure revealed a **retrieval** bug, not a gate failure: the context assembler truncated an unordered set, occasionally dropping the highest-ranked lexical block (the exact penalty clause). A relevance-ordered merge raised real-document recall from 67% to 83% at unchanged false positives. **Lesson:** attribute failures to retrieval vs. extraction before blaming the gate.

5 Recommendations: a checklist for evaluating abstention

1. **Repeat and bootstrap.** Report false positives as mean $\pm$ std over k runs; a single “0%” on a rare event is an illusion.
2. **Establish difficulty empirically.** Cover cases where a similar-but-wrong concept is present, but treat an absence as “hard” only where the baseline actually leaks—adjacency does not imply difficulty; report easy vs. hard.
3. **Annotate by concept, not keyword.** Confirm absence by exhausting paraphrases; prefer a second annotator on real text.
4. **Sweep the vote threshold.** “Answer if any” inflates false positives; report the (k, t) operating curve.
5. **Attribute failures.** Separate retrieval misses from gate decisions before drawing conclusions.

6 Discussion and Limitations

Small, partly synthetic benchmark: a $\sim 1\text{--}2\%$ false-positive rate cannot be estimated tightly from 13–17 ABSTAIN items. **Real-doc external validity is open:** the real ABSTAIN cases are easy; hard real cases coincide with the ambiguous ones awaiting adjudication—the priority extension. **Single annotator; proprietary LLM:** no inter-annotator agreement yet; open-model replication is future work. The system is an **architectural pattern**, not a new ML method; the weight is on the evaluation lessons and released artifacts.

7 Conclusion

An ontology-gated auditor does cut false-positive divergences $\sim 20\text{--}30\times$ on a synthetic benchmark. But the more durable contribution is a caution: evaluating abstention in document audit is easy to get wrong. A rare false-positive rate looks like zero until you bootstrap; a naive baseline ties an abstaining system not only on wholly-missing concepts but on taxonomically-adjacent ones too—so “hard” is the model’s to define, not the taxonomy’s, and a gate helps only where the model actually conflates a pair; and labeling absence by keyword silently corrupts a benchmark on real legal text. We hope the checklist, benchmark, and harnesses help others measure abstention honestly. Future work: hard real-document absences, a second annotator, and open-model replication.

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